Capture-Complete Learning for Transferable Bullet-Hell Control

Touhou Solver Research Project

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Abstract

We study control of the original Japanese Touhou 6 version 1.02h under Wine, with the scientific goal of completing a Lunatic no-miss, no-bomb route and eventually transferring the learning protocol to Touhou 8. The system is capture-complete but prediction-incomplete: it records coherent physical facts, filters only actions that intersect already-instantiated hazards over a fixed short horizon, and leaves action-relative future risk to a learned scorer. We separate the goal metric, no-miss completion probability, from the first optimization target, expected undiscounted physical HIT count over complete episodes. The latter is a dense surrogate and is not claimed to rank policies identically to the goal metric. The initial learner is deliberately limited to a causal decision-epoch dataset, a transparent supervised baseline, and immutable Wine evaluation. History encoders, offline value learning, uncertainty, adaptive collection, learned dynamics, and exact-prefix branching are admitted one at a time only after a simpler preregistered experiment exposes a concrete failure. The paper reports an audited six-stage infrastructure baseline with 108 HITs and a negative serial/parallel equivalence gate. No model has been trained and no learned candidate has been evaluated online.

1 Motivation and Scope

Bullet-hell control combines dense continuous geometry, discrete input, delayed input pickup, partially observed pattern state, long episodes, and sparse but unambiguous failure events. A tempting engineering strategy is to combine a large object encoder, recurrent policy, offline value learner, ensemble uncertainty estimator, active collector, and learned world model at the outset. Such a stack cannot reveal whether a failure came from factual capture, causal action linkage, representation, value learning, exploration, or online inference.

Touhou-Offline instead adopts a *capture-complete, prediction-incomplete* boundary. The adapter records what physically exists at each coherent root, while learning is responsible for regularities not encoded by the minimal observed-hazard filter. The immediate engineering target is a complete Touhou 6 Lunatic route with HIT continuation and zero Bomb. The scientific target is Lunatic NMNB. Touhou 8 is reserved as a transfer test rather than a source of training-time identifiers.

This is a working protocol, not a result paper. In particular, we do not yet claim that the observed-hazard shield prevents all HITs, that object history is sufficient to infer unknown births, that offline RL improves the behavior policy, that a fixed corpus is sufficient, that exact-prefix branching is possible, that accelerated or parallel Wine is evidence-equivalent, or that the method transfers to Touhou 8.

1.1 Frozen outcome and non-goals

The required outcome is one immutable policy that reduces physical HIT count on complete normal-speed original-Wine routes without Bomb, infrastructure failure, forbidden actor input, or widening the observed-hazard action set. Every promoted result must also report NMNB completion rate. The first result need not complete NMNB; it must establish a correctly attributed improvement.

The initial baseline does not attempt online ECL interpretation, global safety proof, exact future-birth prediction, stage or spell routing, policy updates during a run, parallel collection, counterfactual simulation, or a novelty claim. These are non-goals unless a named experiment later earns one of them.

2 Objective and Causal Unit

For a complete physical episode τ, define

$$c_t = \mathbf{1}[\text{a physical life loss is observed on transition } t], \qquad C(\tau) = \sum_{t=0}^{T-1} c_t. \tag{1}$$

Bomb is forbidden and infrastructure failures are reported separately rather than converted into reward. Clearance, graze, score, power, rank, route progress, stage identity, and source identities are not reward.

On the declared evaluation population of complete, infrastructure-valid, zero-Bomb routes, the scientific goal metric is

$$J_{\mathrm{NMNB}}(\pi) = \Pr_{\pi}[C(\tau) = 0]. \tag{2}$$

The first optimization target is

$$J_{\mathrm{HIT}}(\pi) = \mathbb{E}_{\pi}[C(\tau)], \qquad \gamma = 1, \tag{3}$$

with terminal value zero only at the declared physical episode terminal. The two objectives are not interchangeable: equal expected HIT count can hide very different NMNB probabilities. Nevertheless, because C is a nonnegative integer,

$$1 - J_{\mathrm{NMNB}}(\pi) = \Pr_{\pi}[C(\tau) \geq 1] \leq \mathbb{E}_{\pi}[C(\tau)] \tag{4}$$

on that population. HIT count therefore supplies a dense initial surrogate whose claim is exactly "fewer expected HITs," not "higher NMNB probability." Incomplete or invalid trials are never removed silently: their rates and causes are a separate promotion gate. A survival or distributional objective is a separate candidate experiment and may not be introduced silently.

The raw corpus remains frame-linked, but the policy intervention is not every raw frame. A command can remain in an input lease until Wine witnesses its pickup. A derived learner row must therefore join one policy invocation to the next or to the physical episode terminal and contain the published command, its complete behavior probability, witnessed sampled/executed actions, elapsed game frames, accumulated physical HIT cost, the next decision root or terminal, and all exclusion reasons. This decision-epoch view is a causal projection of factual Wine rows, not a simulated successor or a hand-designed gameplay option.

For reward-maximizing implementations the only task reward is $r_k = -C_k$, so maximizing $\mathbb{E}[\sum_k r_k]$ is exactly minimizing J_{HIT}; a cost-form implementation may minimize C_k directly. There is no discount, stage-boundary terminal, first-HIT terminal, or shaped progress term hidden in either convention.

Post-HIT roots remain in the complete episode and in the primary HIT-count view. Separate pre-first-HIT and time-since-HIT views may measure survival-conditioned coverage or train auxiliary factual predictors, but the initial task learner may not relabel the first HIT as the episode terminal.

3 Research Questions

RQ1: portable representation. After exact decision-epoch reconstruction, do current portable physical facts predict factual behavior and future HIT risk on held-out complete episodes, and does a short physical/action history improve them without privileged identity?

RQ2: minimal policy improvement. On one frozen representation and corpus, does the smallest justified offline value learner reduce complete-route expected HIT count relative to the behavior and behavior-cloning baselines under original Wine?

RQ3: coverage and interaction. If the fixed-corpus learner fails, is the demonstrated cause inadequate action/state support, survival-conditioned occupancy, representation, or value estimation; and does one frozen-policy batch of additional Wine data resolve that named failure more efficiently than uniform collection?

RQ4: transfer. Can the episode schema, causal learner view, feature contract, objective, exporter, and evaluator remain unchanged when only the Touhou 6 environment adapter and configuration are replaced by a Touhou 8 adapter?

The source-order question is no longer an open architecture choice. TH06 permits a hazard born after one paused root to enter collision processing before the next completed root. Linking and predicting this class in Wine remains an experiment, but it cannot justify an online birth envelope or ECL interpreter.

4 System Boundary

The system has three independently testable components.

1. A Wine adapter pauses the exact process and records a coherent physical root, input delivery, behavior probability, outcomes, lifecycle, and provenance.
2. A native shield filters actions whose bounded paths overlap hazards already instantiated at that root.
3. An offline learner consumes complete generic episodes and exports a frozen scorer. The scorer may rank the shield set but may never add an action to it.

There is no mandatory online ECL interpreter, stage route, spell rule, boss rule, frame window, coordinate target, or pattern exception. Extracted source scripts may support forensic analysis of the immediate-birth boundary, but source identities are not actor inputs and their successful decoding is not a corpus-admission condition. HIT is an observed cost and does not terminate data collection; Bomb is forbidden.

5 Minimal Online Shield

At decision root t, let x_t be player state, A the 18 directional/focus actions, and O_t the instantiated bullets, lasers, and enemy collision bodies. The adapter lowers O_t to per-frame hazard primitives for a fixed horizon $H = 4$. For each candidate $a \in A$, the native kernel advances the player under the measured movement constants and bounded input-delivery paths. It returns

$$A_{\text{shield}}(x_t, O_t) = \{a \in A : d(x^a_{t+k}, O_{t,k}) > m,\ k = 1, \ldots, H\}, \tag{5}$$

where $m = 0.35$ pixels is the fixed additional collision margin and d uses the source collision geometry.

The claim is deliberately narrow: a returned action is clear of the stored projection of hazards observed at root t. The shield makes no claim about a hazard born after t. The source-order audit at TH06 commit `cc475a0bc3fef38683b0f02224c87ddba0a021d9` establishes this boundary: Player update priority 7 precedes EnemyManager priority 9 and BulletManager priority 11; enemy execution may instantiate a hazard which BulletManager can test for collision in that same update. The audited definitions are in `src/ChainPriorities.hpp`, `src/EnemyManager.cpp`, and `src/BulletManager.cpp` of that source revision. Such births are ordinary partially observed learning outcomes, not justification for a title-specific online predictor. Empty shield sets are recorded as control dead ends; they are not silently widened.

The online path therefore contains only coherent capture, observed-object kinematics, native collision filtering, frozen-policy scoring, and input publication. A retired hand-written long-horizon beam planner is excluded so that gameplay changes can be attributed to the learned policy rather than an untracked rule system.

6 Algorithm-Independent Episode Data

An episode is an immutable sequence

$$\tau_i = \{(o_t, A^{\text{shield}}_t, a_t, \mu_t, o_{t+1}, c_t, b_t)\}_{t=0}^{T_i-1}, \tag{6}$$

where o_t contains factual physical state, a_t distinguishes proposed, published, sampled, and physically witnessed execution, $\mu_t(a)$ is the complete behavior distribution over the admissible actions, c_t is the physical HIT cost, and b_t records resource deltas, lifecycle boundaries, and infrastructure exclusions. Frames and transitions are joined by immutable references and shard hashes. A complete episode has exactly one fewer transition than frame.

The learner derives decision epochs k without changing the raw episode:

$$d_k = (h_k, u_k, \mu_k, \Delta_k, C_k, h_{k+1}, e_k). \tag{7}$$

Here h_k is a versioned portable history ending at a root where the immutable policy was invoked, u_k is the published command or $\perp$ when publication did not occur, Δ_k is elapsed game frames to the next invocation or the episode terminal, C_k is the sum of factual HIT indicators over that interval, and e_k contains publication, pickup, observation-gap, and learning-exclusion evidence. A $\perp$ row is retained for accounting but is not an action-conditioned training sample. Commanded, sampled, and executed actions remain distinct. The decision rows must conserve complete-episode HIT count exactly.

The canonical raw layer retains measured player state; occupied bullet, laser, enemy, player-shot, and item records; visual and collision geometry; resource counters; input state; clocks; capture retries; and executable, native, policy, and schema provenance. Derived tensors are versioned products rather than replacements for raw shards. This permits multiple feature builders and RL algorithms to consume identical episode identities.

The same admitted episode may expose three non-authoritative derived views:

1. the complete decision-epoch HIT-cost view used by the initial task;
2. a pre-first-HIT or time-since-HIT diagnostic view used only to measure survival-conditioned coverage and factual prediction;
3. a complete physical view, including post-HIT roots, used for auxiliary representation and dynamics targets.

Every view names the same episode and raw hashes. No view may invent a next state, turn an unexecuted action into a transition, or admit a diagnostic stop-on-HIT run as training evidence.

Deployable features may contain normalized player state, unordered physical object primitives, action history, declared portable resources, and the shield mask. They may not contain stage, boss, spell, ECL, RNG, memory address, run identifier, future outcome, or post-treatment facts. Unordered hazard sets motivate permutation-invariant encoders such as Deep Sets [4], but this complexity is admitted only after a transparent vector baseline passes.

7 Minimal Learning Architecture

The strongest simple baseline has exactly three independently changeable parts:

1. a deterministic builder that converts complete factual episodes into audited decision rows and whole-episode splits;
2. a transparent current-observation behavior-cloning scorer,
$$\mathcal{L}_{\mathrm{BC}}(\theta) = -\sum_{i,k} \log \pi_\theta(u_k \mid h_k, A_k^{\mathrm{shield}}); \tag{8}$$
3. an immutable exporter that reproduces the offline action distribution bit-for-bit, ranks only observed-shield-admissible actions, and is evaluated on complete original-Wine episodes.

Action frequency and the existing reactive policy are non-fitted controls. BC is a plumbing and representation test, not a policy-improvement claim. The first known constraint is that the current online policy API exposes scalar player facts, object counts, and shield summaries but not the physical object history contemplated by RQ1. The baseline intentionally uses that smaller surface first. An API expansion is earned only after an offline held-out experiment shows that a precisely specified additional portable history improves a factual target and after offline/online feature parity is testable.

8 Candidate Ladder and Admission Gates

Only one rung may change at a time. Every rung freezes its corpus inventory, whole-episode split, feature schema, target schema, seeds, acceptance statistic, and artifact hashes before fitting.

L0: causal data audit (active next step). Build and test decision-epoch rows. Admission requires exact raw linkage, command/sample/execution separation, propensity normalization, interval and episode HIT conservation, exclusion replay, forbidden-feature rejection, and two materially different feature builders reading the same episode identities. No model result is meaningful before L0 passes.

L1: transparent supervised baseline. After L0, collect the initial training inventory serially, with unchanged retail clock/update semantics, coherent capture suspension, the frozen declared 20% uniform observed-shield mixture, and complete episodes; this is non-adaptive baseline collection, not L5. Freeze the episode count and split before fitting. Report per-admissible-mask action frequency, the reactive behavior baseline, and current-observation BC. BC passes the learnability test only if its held-out complete-episode negative log likelihood improves over action frequency with a preregistered episode-bootstrap interval and its calibration does not regress beyond a frozen tolerance. Accuracy alone is insufficient.

L2: one representation change. Add a short, fixed physical/action history and simple factual heads for HIT within fixed horizons and observed-hazard-set shrinkage or empty-set events. History is retained only if it improves held-out log loss or another frozen proper score over L1 with an episode-level interval. Deep Sets [4] are considered only if a transparent bounded vector summary fails; a recurrent encoder is considered only if history has already shown value. Auxiliary heads remain separate from reward.

L3: one offline value method. Only after L1–L2 establish a usable representation may BC be compared with one expected-HIT value learner on the identical data. IQL is the first candidate because its critic avoids directly evaluating unseen actions and its actor remains advantage-weighted behavior cloning [2]. CQL is not run in parallel; it is admitted only if the IQL experiment identifies unsupported-action overestimation that its conservatism is designed to address [3]. Offline metrics can reject a candidate but cannot promote it without complete Wine evaluation.

L4: objective alternatives. A bounded survival or distributional HIT model is admitted only when the corpus contains enough late no-HIT prefixes or zero-HIT episodes to avoid a degenerate all-failure target, or when policies with similar expected HIT count show a measured difference in NMNB probability. Until then, first-HIT survival is a diagnostic, not the task terminal or deployment score.

L5: interaction and uncertainty. Frozen-policy batch collection is admitted only after L3 demonstrates either policy improvement or a measured support failure. The first additional batch comparison repeats the declared uniform-shield mixture at a fixed interaction budget. Ensemble uncertainty, active collection, OPE, or baseline bootstrapping may each be tested later against one named support or calibration failure. A positive action-probability floor does not by itself establish state coverage or reliable long-horizon OPE.

L6: learned dynamics and exact-prefix branching. Learned birth models or risk-aware MPC are considered only if model-free value learning fails while held-out future-hazard prediction succeeds. Predictions remain scores and never extend the shield certificate. Real-Wine branching is considered only after a serial same-seed, same-policy, exact-action-prefix gate reaches a target root with identical factual and input-delivery digests. The negative parallel gate does not establish this ability. Every training branch must still be a complete admitted Wine episode.

Decision Transformer [1], large recurrent world models, simultaneous IQL/CQL sweeps, and the proposed SABER-RL composite (survival critic, recurrent belief, ensemble uncertainty, active collection, and Wine branching) are not active candidates. SABER-RL is decomposed across L2 and L4–L6; neither the name nor the combination is a novelty claim. These methods add independent failure modes before the baseline has exposed a constraint they solve.

8.1 Algorithm and policy record

Policy or method	Status	Evidence or admission condition
Reactive scorer	Executed control	E0 completed one route with 108 HITs; this is infrastructure evidence, not a learned-policy result.
20% uniform-shield mixture	Executed diagnostic	E5 Stage 4 runs had 16, 10, and 13 HITs, but factual equivalence failed; no training inventory or policy claim was admitted.
Action frequency	Not run	First non-fitted L1 comparator after L0 and initial serial inventory.
Current-observation BC	Not run	First fitted candidate; requires the L1 held-out likelihood and calibration gate.
Short-history BC and factual risk probes	Not run	One L2 representation ablation after current-observation BC.
IQL	Not run	First L3 value candidate after representation learnability.
CQL	Not run	Admitted only after measured IQL unsupported-action overestimation.
Survival/distributional value	Not run	Admitted only by L4 coverage or objective evidence.
Deep Sets, recurrence, ensembles, active collection, learned MPC, branching	Not run	Each requires its distinct L2, L5, or L6 gate; none is active architecture.
SABER-RL composite	Not admitted	Its survival, recurrence, uncertainty, collection, and branching claims must each pass their separate rung before a combined-method experiment exists.

Table 1: Algorithm record at this checkpoint. "Not run" means no fit and no online learned candidate, not an unreported negative result.

9 Experimental Protocol

Every result must name the Git commit, executable hash, policy hash, schema, full command, complete episode IDs, and artifact paths. Splits are by complete episode. Negative results are retained. Infrastructure failures are reported separately and never counted as gameplay improvements. Before a fit, the experiment must additionally freeze its corpus inventory, split-group hashes, decision-row schema, feature and target schemas, seeds, comparator, acceptance statistic, and stopping rule. A post-hoc metric may be reported but cannot promote the candidate.

ID	Decisive question	Status
E0	Does the reset runner complete Practice 4–6 and one full route?	Completed
E1	Can a post-root birth hit before the next controllable root?	In progress
E2	Does the causal learner view preserve actions, propensities, and HITs?	In progress
E3	Do portable supervised baselines generalize by episode?	Not run
E4	Does one offline value learner reduce complete-route HIT cost?	Not run
E5	Does shared-host parallel collection preserve exact facts?	Negative result
E6	Does a new frozen Wine batch resolve a measured coverage failure?	Not run
E7	Which surfaces change under a Touhou 8 adapter?	Not run

Table 2: Preregistered experiment ledger. Status changes require immutable artifacts.

9.1 E0: infrastructure baseline

Run natural Lunatic Practice Stages 4, 5, and 6 and then one natural six-stage route on original Wine with unchanged retail update semantics, coherent capture suspension, and the frozen baseline policy. HIT continuation and zero Bomb are mandatory. Accept only if every declared run completes, frames equal transitions plus one, no record is dropped, player successors and stored shield decisions replay exactly, every HIT has linked before/action/after evidence, and cleanup leaves no worker process. HIT count is measured, not gated.

At solver commit `af9900524520b72934a4c55e2f44118f88094633`, all three bounded Practice runs completed under isolated four-CPU workers after the atomic-input cleanup. Table 3 reports their complete audits. Every declared frame and transition loaded, every Bomb/callback/integrity/scope count was zero, all checked player successors were bit-exact, and every stored dense shield sample reproduced with no unsafe or conservative divergence. The higher Practice observation-gap rates motivated eight CPUs for the canonical route and future evidence collection; gap transitions remain explicit and are excluded rather than fabricated.

Stage	Frames	Transitions	HIT	Gap %	Successors	Shield
4	25,564	25,563	18	0.536	24,344	58
5	23,129	23,128	23	0.968	19,503	64
6	30,981	30,980	34	0.507	28,763	64

Table 3: Canonical E0 Practice evidence. Successor and shield columns are checked counts; every mismatch/divergence count is zero.

The Stage 4–6 run IDs are, respectively, `20260821T080408Z-471737400`, `20260821T081924Z-372154700`, and `20260821T080532Z-100425500`; their audit SHA-256 digests are `c4178c3c9e5b7d88f7a950d8fe18403630a8524320ae72dc0d0ce97c0a26bb2d`, `2677f2b9689c42c5483a2e278d077a6b721623430d89f6c39a41bc92b3c34ebd`, and `e34bec2844dc7ce92f572326dd262a1e7f718d6798bc9a6c9dd7124692308c17`.

The natural-RNG full route then reached Ending with run ID `20260821T083303Z-499458600`: 135,561 frames, 135,560 transitions, 108 physical HITs, zero Bomb, zero dropped record, and zero capture, corpus, policy, trace, or other infrastructure failure. All six Stages were present. The audit classified all 108 HITs as observed-shield-empty, reproduced 64 dense shield samples without divergence, and matched 126,939 eligible player successors bit-for-bit. Only 28 links were excluded for observation gaps (0.0207%); capture and solve p99 were 8.044 and 0.918 ms respectively. The independent baseline verifier passed every declared check. The immutable SHA-256 digests for report, audit, run metadata, and manifest are respectively `f2577d31ca5a51108911b4d60a957a7e`

`2e891edda5ba337fbca1f70386e37686`, `8d13d3196261f4f8710d8187cdec898e364b67f024696a8cca1991aff467baa9`, `9f9fbdd75dbe89651c789dc502dab6a50323a93ca488051720538308079e3396`, and `8423bae1ef93decd50605a5e47900ee0ded3a6096d6067fb4c0a8eca944ec1d8`. E0 is therefore complete; the 108-HIT value is a baseline measurement, not a policy-quality claim.

9.2 E1: immediate-birth boundary

Use source update order and adjacent physical roots around every HIT and newly observed overlapping object. Source order already proves that post-root birth and same-update collision are possible. The remaining experiment links this class to adjacent Wine roots and measures whether short physical history predicts its action-relative cost. No source/ECL envelope is added to the online shield; pattern-specific exceptions are forbidden.

9.3 E2: causal learner view

The existing generic loader already admits complete E0 episodes and enforces frame/transition linkage. E2 is not complete until it also removes optional source/ECL material, derives decision epochs from policy invocations and input leases, reproduces proposed/published/sampled/executed action sequences, conserves interval and episode HIT totals, validates complete propensities and exclusions, and lets two different feature builders consume the same raw episode identities. A feature-parity fixture must prove that the online exporter receives exactly the versioned values produced by offline replay. E2 changes no policy and collects no new gameplay evidence.

9.4 E3: supervised representation baselines

After E2, collect the initial inventory serially with unchanged retail clock/update semantics, coherent capture suspension, and the frozen 20% uniform observed-shield mixture; its episode count, seeds, and stopping rule are declared before the first run. Then freeze its whole-episode train/validation split and compare the per-admissible-mask action-frequency control with current-observation BC. Only if the preregistered held-out negative-log-likelihood and calibration gate passes may one short fixed history be added. The same split may then support simple executed-action factual probes for HIT within fixed horizons and future observed-shield collapse. These probes test information content; they do not define reward, infer unexecuted successors, or constitute policy improvement.

9.5 E4: minimal offline policy improvement

E4 begins only after E3 identifies one frozen portable representation. It compares BC with one offline expected-HIT method, initially IQL, on the same decision rows. Offline validation reports proper cost-prediction scores, calibration, action support, and episode-bootstrap intervals. Promotion then requires normal-speed original-Wine complete routes, zero Bomb, passing infrastructure contracts, lower HIT count under a preregistered serial comparison, and reported NMNB rate. Exact paired claims are allowed only if a separate exact-prefix replay gate passes; otherwise independent complete-route episode intervals are used. No offline estimate alone can promote a scorer.

9.6 E5: collection equivalence

Parallel workers are enabled only after E0. Serial and parallel fixed-seed runs must first agree on factual/action traces before independent natural-RNG collection begins. The first declared collection policy is an immutable 20% mixture of uniform exploration over the observed-shield set and the generic reactive baseline, with complete propensities and independent policy RNG streams per episode. Clock acceleration is a distinct hypothesis and cannot produce evidence until it passes an unchanged-retail-clock differential contract.

At commit 76782a4f37e0d12a9a2384561b53e68ceaf998ae, the first declared shared-host test used two workers with eight logical CPUs each, disjoint game/controller partitions, retail frame multiplier 1.0 with coherent capture suspension, fixed retail seed 0x1234, and the immutable 20% uniform-shield policy. The outer process tree used SCHED_OTHER nice 19 and idle I/O priority so normal-priority background proof jobs retained scheduling precedence. The exact command was:

```
ionice -c 3 nice -n 19 taskset -c 32-47 \
  .venv/bin/python scripts/gate_parallel_wine.py \
  --pool reference/wine-workers-v2-zklean-safe/pool.json \
  --policy-plugin src/th06_rl/policies/uniform_shield_exploration.py \
  --policy-state config/uniform_shield_exploration.json \
  --practice-stage 4 --diagnostic-rng-seed 0x1234 \
  --artifact-root artifacts/parallel-gate-exp02-76782a4-zklean-safe-nice19 \
  --corpus-root corpora/parallel-gate-exp02-76782a4-zklean-safe-nice19 \
  --output artifacts/parallel-gate-exp02-76782a4-zklean-safe-nice19/gate.json
```

The retail executable, native kernel, policy plugin, policy state, and worker pool SHA-256 digests were respectively 9f76483c46256804792399296619c1274363c31cd8f1775fafb55106fb852245, d26439f01400121a99137ff7d411b4bb6a2ff480daaaf1195e85a1b6cb671673, c70a2ad0a03bb189a01baabb62f6f404f6c4c91aa0e16c86645bfc0925f3e805, a63fbbb0b6d5886d327eb28b4b3d683cc13c6c759bffde8f9b2f554649068ef1, and 76fe8c634855a2a9f11f4a5b83ab0c3498eb914df384ed40f206f199750447ad. The pool and audit schemas were th06-rl-normal-speed-wine-pool-v2 and th06-rl-infra-audit-v2.

Run	Frames	Transitions	HIT	Gaps (%)	Successors	Shield
Serial 0	25,106	25,105	16	0.0159	24,125	62
Concurrent 0	23,154	23,153	10	0.0346	22,535	64
Concurrent 1	25,455	25,454	13	0.1729	24,617	64

Table 4: E5 fixed-seed differential. Every row individually completed with zero Bomb, successor mismatch, unsafe divergence, conservative divergence, drop, callback failure, integrity error, or leftover process.

The run IDs were 20260821T101125Z-298490100, 20260821T102110Z-520526700, and 20260821T102110Z-450384400; their audit SHA-256 digests were 5393077138d1a769e12a703e35e3cae11201e79859529acfcf52b40231153366, 8a07cabb6ca93b897853c79ede1fc7cd24151f0a356b8390981cb2d498ba8aaf, and 8bfaff06bc25ec808f258cc0e5d1b576627741e5b53928461b639fdd43035e4d. Capture p99 was 8.420, 8.340, and 8.692 ms; solve p99 was 0.815, 0.784, and 0.895 ms. Thus every run passed its individual latency and semantic checks, but both HIT count and normalized factual digest failed exact equality. The gate rejected concurrent-worker-0 and emitted no admission ledger. This is a negative E5 result, not permission to weaken equality: parallel collection remains disabled.

An earlier SCHED_IDLE attempt is retained as a scheduling negative: its serial run completed with zero semantic mismatch but 883 observation gaps (3.62%), above the unchanged 0.5% limit. Its audit digest is 00a5a1f8fab26fac909c4f0d421ba9ece13c418b25bc81ccdefdfc9c2a68da12.

9.7 E6: frozen-batch interaction

E6 is admitted only if E4 records a specific coverage failure or an improved candidate whose remaining error is support-limited. One immutable policy is used for each complete Wine run; parameters never update or reload within a run. The first additional collection comparator is the declared 20% uniform observed-shield mixture. Any uncertainty- or novelty-guided collector is a separate later arm with a minimum action-probability floor and the identical Wine interaction budget. All episodes remain complete, immutable, and linked to their behavior distributions. E6 was not run.

9.8 E7: transfer

E7 replaces only the environment adapter and configuration and reports unchanged versus changed code/schema surfaces, zero-shot behavior, and fine-tuning sample cost. It was not run.

This documentation checkpoint follows E0 and the negative E5. The only active next implementation is E2/L0 causal learner-view work. No new corpus collection, model fit, policy-API expansion, or learned online candidate has yet been authorized as evidence.

10 Threats to Validity

The shield horizon may be too short, and unknown births may violate its narrow claim. Expected HIT count is only a surrogate for NMNB probability and may rank nonzero-HIT policies differently. HIT continuation changes later physical occupancy through lifecycle, resources, rank, bullet clearing, and invulnerability; complete-route training and survival-conditioned diagnostics must therefore be reported separately. Behavior action support does not guarantee support for histories reached by a stronger no-HIT policy, and a positive propensity floor does not prevent long-horizon OPE variance.

Complete-route episodes may be too few for stable episode-level intervals, future-HIT prediction, or uncertainty estimates. Wine scheduling may introduce observation gaps; command pickup may be mistaken for a new policy decision; and Touhou 6–specific memory facts may leak through an apparently generic feature builder. E0 and E2 are therefore prerequisites, not housekeeping. The transfer claim is falsified if downstream learner, corpus, objective, or evaluator semantics require title-specific branches.

11 Artifact and Reproducibility Contract

The implementation uses repository-relative scripts. The LaTeX artifact is built by `scripts/check_paper.sh`. Runtime setup, native compilation, Wine execution, corpus validation, and route verification have distinct entry points. The canonical compiled paper is tracked at `paper/main.pdf`; intermediate LaTeX files stay in the ignored `paper/build/` directory. Wine prefixes, corpora, and fitted models remain untracked outputs. Committed experiment claims must point to immutable hashes rather than mutable filenames.

12 Conclusion

The project asks how much complete-route HIT reduction can be learned from portable physical history when exact Wine interaction is costly, input pickup is delayed, future births are partially observed, and the online filter covers only instantiated hazards. The goal is NMNB; the first optimization claim is strictly expected undiscounted HIT reduction. The initial learner has three parts:

an audited decision-epoch view, transparent BC, and immutable shield-constrained Wine evaluation. Every additional representation, algorithm, objective, uncertainty mechanism, or collection strategy must solve one measured failure of that baseline.

E0 establishes the serial factual data plane, while E5 rejects the current parallel pool. E2 causal learner reconstruction is the next active step. No model has been fit, no learned policy has run online, exact-prefix branching is not established, and this checkpoint makes no learning, NMNB, or transfer claim.

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